

Characterizing insect communities using multivariate analyses

ENTMLGY 6707 Entomological Techniques and Data Analysis

Learning objectives

- 1) Distinguish among different ordination techniques
- 2) Understand how to apply these methods to your data
- 3) Interpret the outputs of the ordination analyses

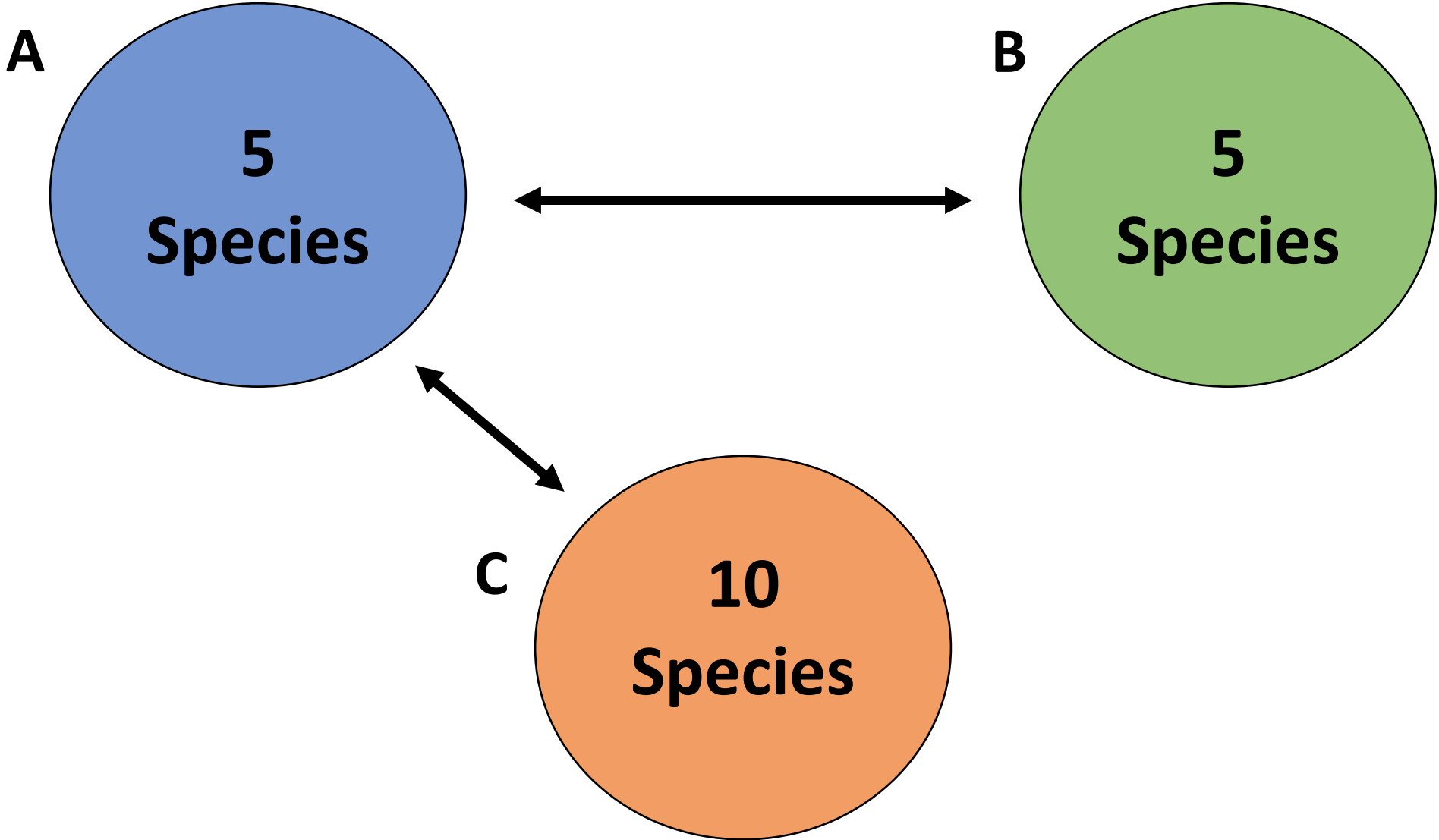
Traditional taxonomic approach to study insect diversity

Measure unique “types” and compare among samples or communities

Type: taxonomic unit (species, genera, families, orders)

1. **Abundance** measures the number of individuals of each distinct type
2. **Richness** measures the number of distinct types
3. **Evenness** measures the relative abundance of each type
4. **Diversity** measures the number of types and their evenness

What information about these communities are we missing?



Advantages of multivariate ordination methods

- Analysis of multiple environmental factors on many species simultaneously
- Represents sample and species relationships in a low-dimensional space along (ideally) important and interpretable environmental gradients
- Capable of handling noisy and redundant data
- Accommodates sparse data (i.e., large portion of the entries consist of zeros) because most species are infrequent

Types of ordination techniques



Indirect gradient analysis (aka unconstrained ordination)

- Utilizes only the species x sample matrix
- Any environmental data are used after the analysis to aid with interpretation

**Nonmetric Multidimensional Scaling and Principal Component Analysis*

Types of ordination techniques



R package
vegan

Indirect gradient analysis (aka unconstrained ordination)

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**Nonmetric Multidimensional Scaling and Principal Component Analysis*

Direct gradient analysis (aka constrained ordination)

- Utilizes environmental data in addition to a species x sample matrix
- Assess whether species composition is related to measured environmental data

**Canonical Correspondence Analysis and Redundancy Analysis*

1. Hemlock girdled
2. Hemlock logged
3. Hemlock control
4. Hardwood control

$n = 2$

90 x 90 m plots



Methods in Ecology and Evolution



Methods in Ecology and Evolution 2010, 1, 168–179

doi: 10.1111/j.2041-210X.2010.00025.x

Experimentally testing the role of foundation species in forests: the Harvard Forest Hemlock Removal Experiment

Aaron M. Ellison*, Audrey A. Barker-Plotkin, David R. Foster and David A. Orwig

Harvard Forest, Harvard University, 324 North Main Street, Petersham, MA 01366, USA



1. Hemlock girdled
2. Hemlock logged
3. Hemlock control
4. Hardwood control

n = 2

90 x 90 m plots



Harvard Forest

Established in 1907

Harvard University's 4000 acre laboratory & classroom

Long Term Ecological Research site since 1988

Harvard Forest Data Archive

HF106

Understory Vegetation in Hemlock Removal Experiment at Harvard Forest since 2003

Related Publications

Data

- [hf106-01](#): species codes ([preview](#))
- [hf106-02](#): o-layer and substrate ([preview](#))
- [hf106-03](#): shrub and herb cover ([preview](#))
- [hf106-04](#): seedling count and cover ([preview](#))
- [hf106-05](#): saplings ([preview](#))
- [hf106-06](#): species list ([preview](#))
- [hf106-07](#): sapling heights ([preview](#))

Environmental data:

Shrub and herbaceous species in the understory



[illegible][illegible]

Nonmetric Multidimensional Scaling (NMDS)

Used to assess differences in species composition among sites, treatments, etc.

Represents (as well as possible) the ordering relationships among sites in species space

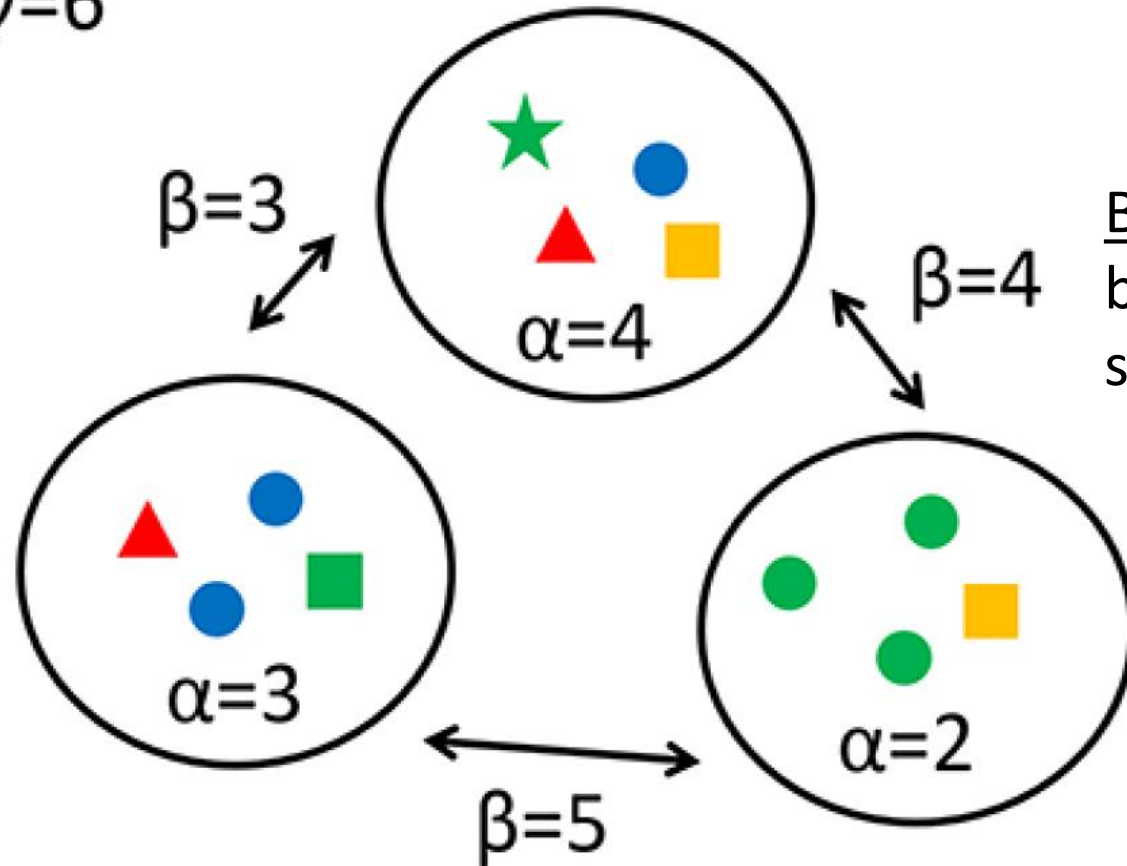
Based on a distance (dissimilarity) matrix as a measure of beta-diversity

Iterative method that maximizes the rank order correlations between the distances in the dissimilarity matrix and the distances in low-dimensional space

Robust technique – no assumptions of normality or linear relationships among variables.

Gamma diversity (γ) = total number of species across all habitats within a landscape

$\gamma=6$



Beta diversity (β) = diversity between habitat patches, samples, or sites.

Alpha diversity (α) = diversity within a particular habitat patch, sample, or site.

Figure 5. Illustration of the concept of alpha, beta, and gamma diversity. Colored symbols represent species, circles represent habitat patches, and the large rectangle represents the landscape.

Beta diversity metrics

Beta diversity (β) = diversity between habitat patches, samples, or sites.

Introduced by R.H. Whittaker in 1960

- “The extent of change in community composition, or degree of community differentiation, in relation to a complex-gradient of environment”

$$\beta_W = (\gamma - \alpha) / \alpha$$

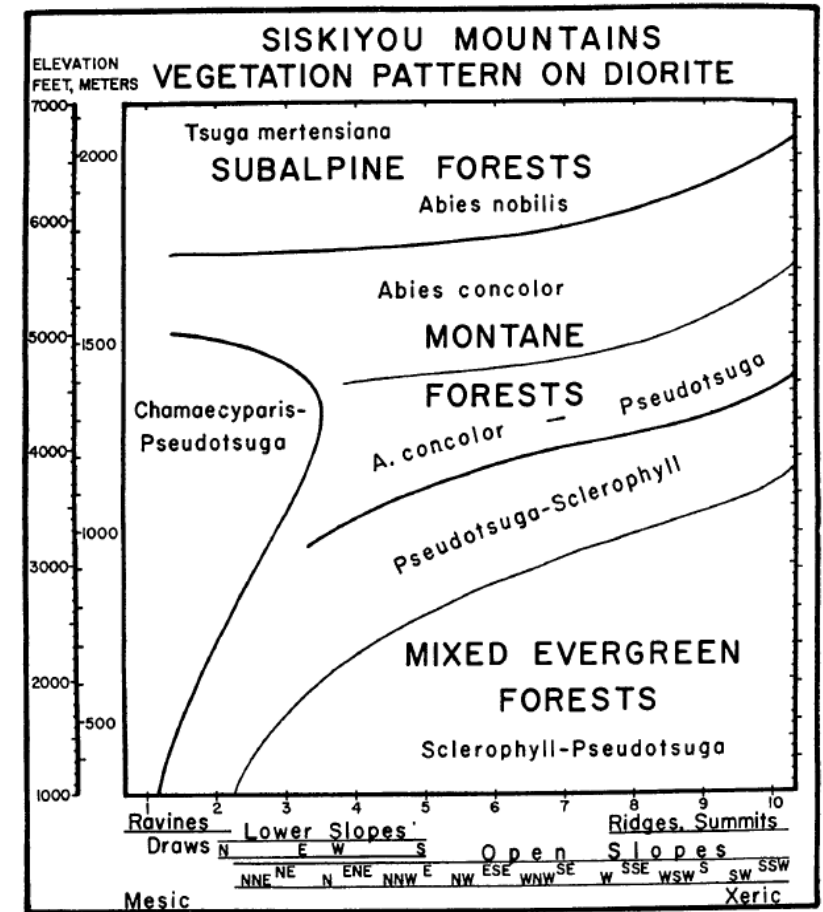


FIG. 11. Mosaic chart of vegetation on quartz diorite, central Siskiyou Mountains, Oregon.

Measuring the compositional (dis)similarity among communities

Incidence-based or abundance-based dissimilarity matrix

- Sorensen (Bray-Curtis)
- Jaccard
- Euclidean

Create a matrix that compares the number of shared species to the number of unique species among two sites (pairwise)

Whittaker formula: $\beta_W = (b + c) / (2 * a + b + c)$

a = number of species shared between two samples

b = number of species unique to sample 1

c = number of species unique to sample 2

Jaccard dissimilarity matrix

vegdist()

```
> dis.matrix.pa <- vegdist(ants4[,1:30], method = "jaccard")
> dis.matrix.pa
```

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
2	0.6000000														
3	0.8888889	0.6666667													
4	0.2500000	0.5000000	0.8571429												
5	0.7000000	0.6250000	0.6000000	0.7777778											
6	0.5714286	0.7142857	0.9230769	0.6153846	0.6923077										
7	0.6666667	0.3333333	0.5000000	0.5714286	0.5000000	0.7692308									
8	0.5000000	0.5454545	0.8000000	0.5454545	0.5000000	0.6250000	0.6000000								
9	0.6842105	0.7222222	0.8823529	0.7222222	0.7058824	0.6818182	0.7647059	0.5789474							
10	0.7142857	0.5454545	0.8000000	0.6666667	0.8461538	0.8421053	0.7272727	0.6666667	0.5789474						
11	0.8888889	0.6666667	0.0000000	0.8571429	0.6000000	0.9230769	0.5000000	0.8000000	0.8823529	0.8000000					
12	0.5555556	0.4285714	0.6000000	0.4285714	0.7500000	0.7857143	0.5000000	0.5000000	0.7058824	0.5000000	0.6000000				
13	0.6363636	0.5555556	0.7142857	0.5555556	0.6666667	0.6428571	0.6250000	0.5833333	0.7368421	0.6923077	0.7142857	0.5000000			
14	0.6666667	0.6250000	0.8750000	0.7058824	0.6875000	0.5263158	0.7500000	0.5555556	0.5652174	0.5555556	0.8750000	0.6875000	0.7222222		
15	0.7777778	0.5000000	0.3333333	0.7142857	0.6666667	0.8461538	0.2500000	0.7000000	0.8235294	0.7000000	0.3333333	0.4000000	0.5714286	0.8125000	
16	0.6363636	0.3750000	0.7142857	0.5555556	0.5000000	0.5384615	0.4285714	0.4545455	0.5882353	0.6923077	0.7142857	0.5000000	0.4444444	0.5625000	0.5714286

```
> |
```

Run a non-metric multidimensional scaling (NMDS) model

```
> nmds.ants.pa <- metaMDS(dis.matrix.pa, trymax = 500, autotransform = TRUE, k = 2)
```

```
Run 0 stress 0.1459161
Run 1 stress 0.1401423
... New best solution
... Procrustes: rmse 0.1568134 max resid 0.3461763
Run 2 stress 0.1401423
... New best solution
... Procrustes: rmse 7.163306e-06 max resid 1.843059e-05
... Similar to previous best
Run 3 stress 0.1401423
... Procrustes: rmse 3.603234e-06 max resid 9.665341e-06
... Similar to previous best
Run 4 stress 0.1543155
Run 5 stress 0.1418973
Run 6 stress 0.1454809
Run 7 stress 0.1401423
... New best solution
... Procrustes: rmse 4.744046e-06 max resid 1.131643e-05
... Similar to previous best
Run 8 stress 0.1543156
Run 9 stress 0.1854143
Run 10 stress 0.1513608
Run 11 stress 0.1543156
Run 12 stress 0.145916
Run 13 stress 0.1510867
Run 14 stress 0.1401423
... New best solution
... Procrustes: rmse 5.499765e-06 max resid 1.425497e-05
... Similar to previous best
Run 15 stress 0.1401423
... Procrustes: rmse 1.779235e-06 max resid 3.274561e-06
... Similar to previous best
Run 16 stress 0.1560107
Run 17 stress 0.1560108
Run 18 stress 0.1401423
... Procrustes: rmse 1.11992e-05 max resid 2.74026e-05
... Similar to previous best
Run 19 stress 0.1459161
Run 20 stress 0.1459162
*** Best solution repeated 3 times
```

```
> nmds.ants.pa # stress is quality of fit
```

Call:

```
metaMDS(comm = dis.matrix.pa, k = 2, trymax = 500, autotransform = TRUE)
```

global Multidimensional Scaling using monoMDS

Data: dis.matrix.pa

Distance: jaccard

Dimensions: 2

Stress: 0.1401423

Stress type 1, weak ties

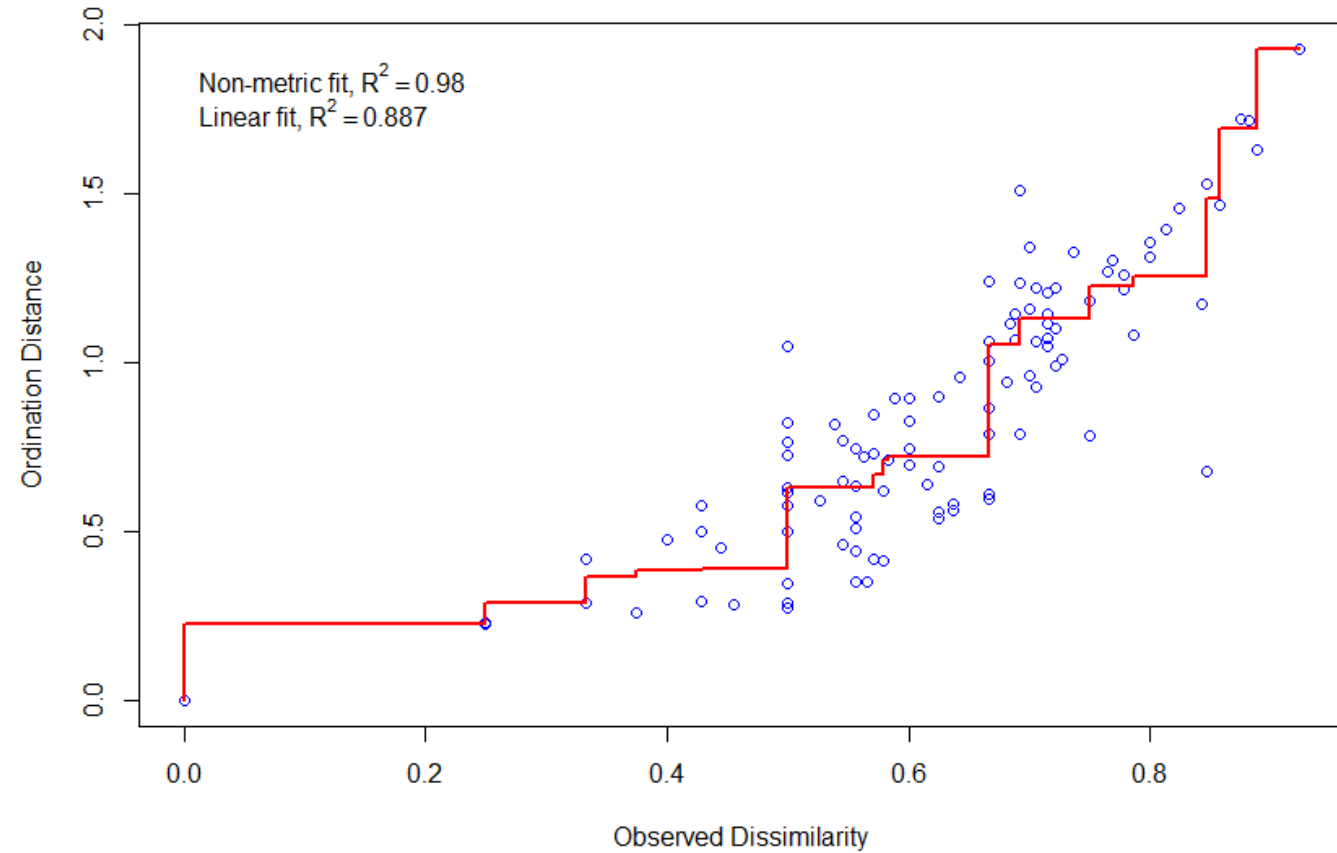
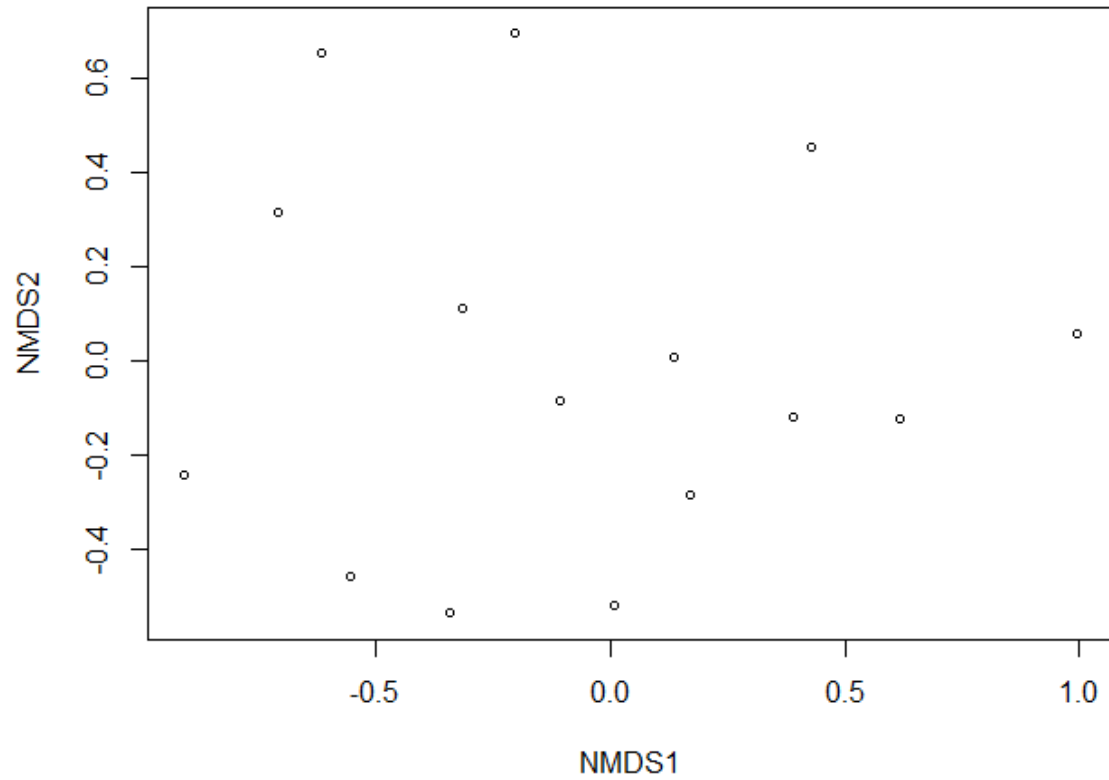
Best solution was repeated 3 times in 20 tries

The best solution was from try 14 (random start)

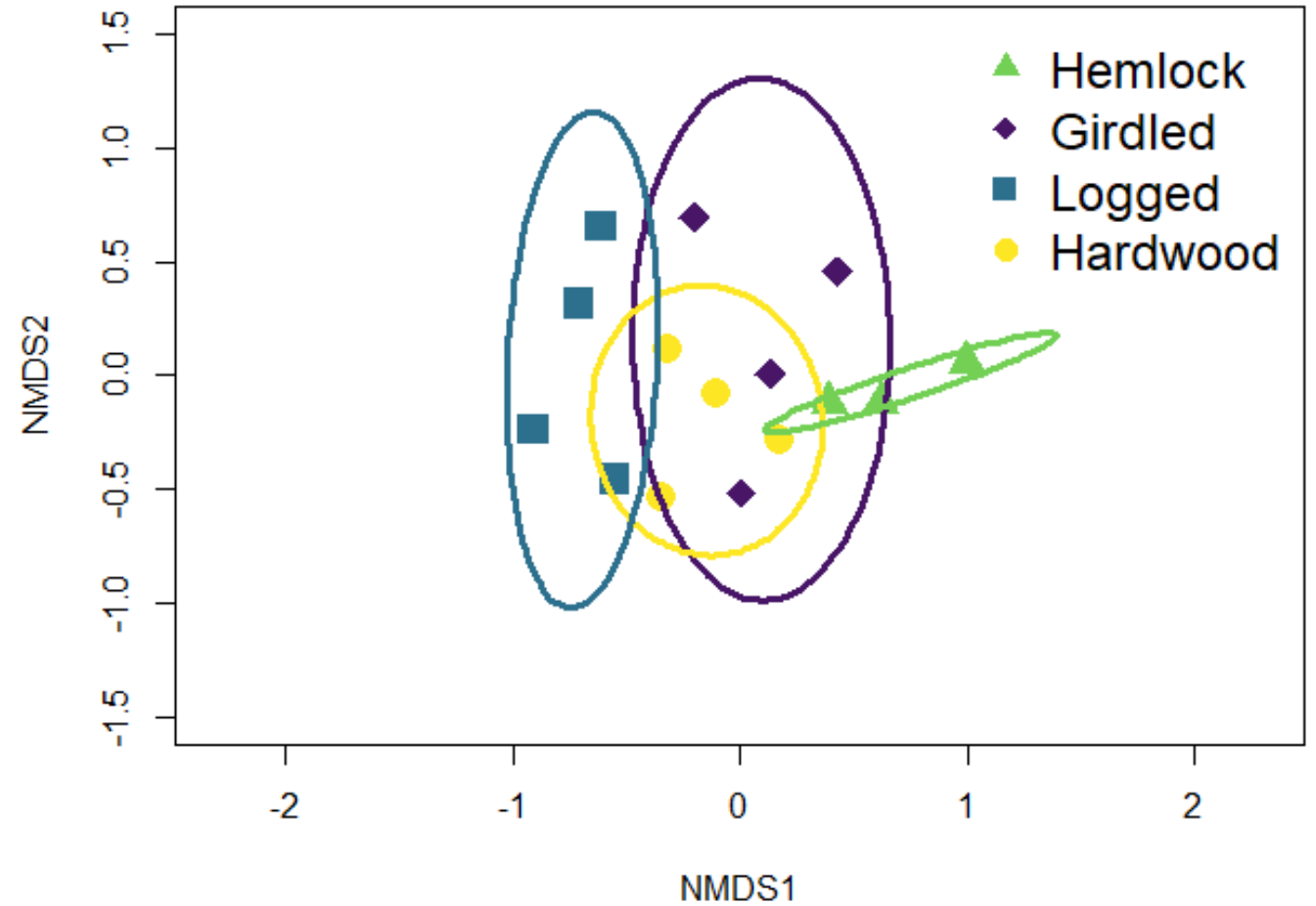
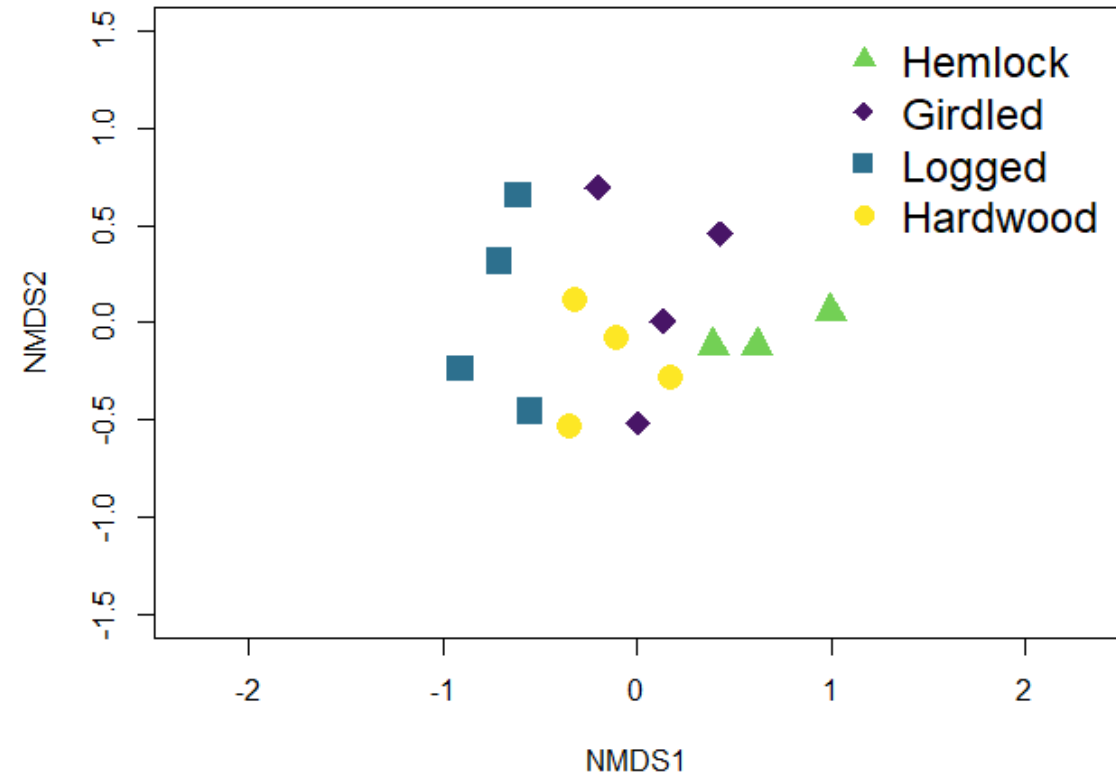
Scaling: centring, PC rotation, halfchange scaling

Species: scores missing

Run a non-metric multidimensional scaling (NMDS) model



Visualize with a non-metric multidimensional scaling (NMDS) plot



NMDS often paired with PERMANOVA and BETADISPER

Permutational multivariate analysis of variance (PERMANOVA)

- tests whether the group centroid of communities differs among a categorical grouping factor in multivariate space

Homogeneity of multivariate group dispersion (BETADISPER)

- tests whether the dispersion of a categorical grouping factor from its spatial medial is different between groups.
- Multivariate analogue of Levene's test for homogeneity of variances

PERMANOVA

```
> adonis2(dis.matrix.pa ~ ants4$treatment, permutations = 999)
```

```
Permutation test for adonis under reduced model
```

```
Terms added sequentially (first to last)
```

```
Permutation: free
```

```
Number of permutations: 999
```

```
adonis2(formula = dis.matrix.pa ~ ants4$treatment, permutations = 999)
```

	Df	SumOfSqs	R2	F	Pr(>F)
ants4\$treatment	3	1.3801	0.43335	3.0591	0.001 ***
Residual	12	1.8045	0.56665		
Total	15	3.1846	1.00000		

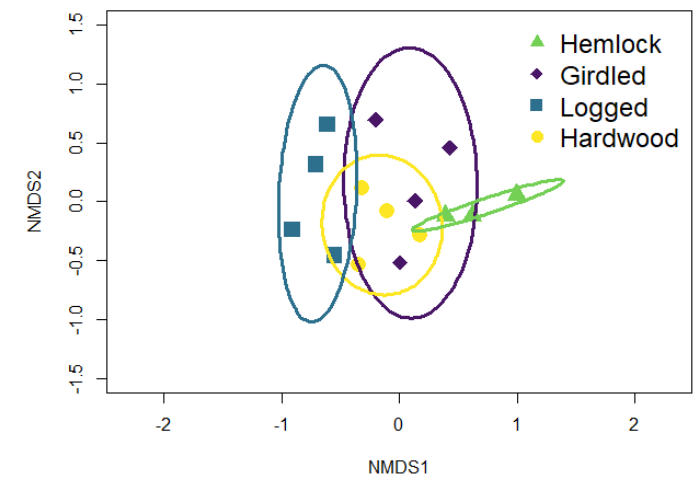
```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

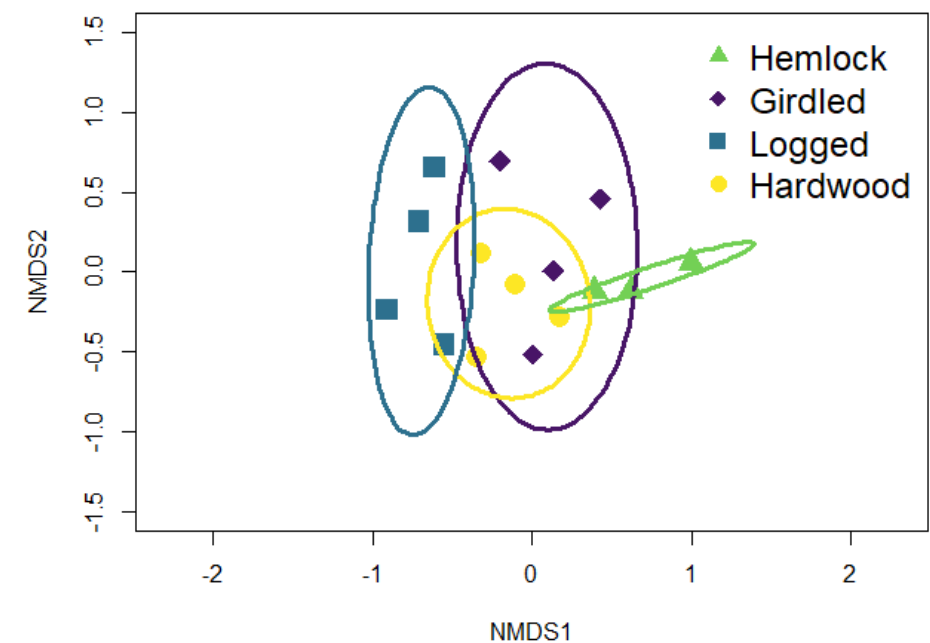
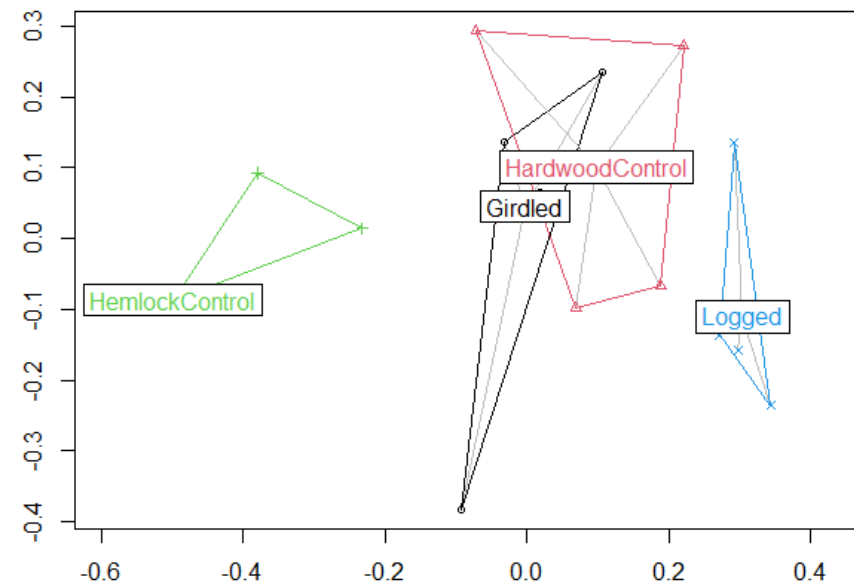
```
> pairwise.adonis(dis.matrix.pa, ants4$treatment)
```

	pairs	Df	SumsOfSqs	F.Model	R2	p.value	p.adjusted	sig
1	Logged vs Girdled	1	0.3164513	1.5383579	0.2040707	0.157	0.942	
2	Logged vs HemlockControl	1	1.0162804	7.9057299	0.5685232	0.032	0.192	
3	Logged vs HardwoodControl	1	0.2753550	1.7404940	0.2248557	0.117	0.702	
4	Girdled vs HemlockControl	1	0.4099642	2.8759151	0.3240134	0.018	0.108	
5	Girdled vs HardwoodControl	1	0.1401176	0.8136623	0.1194163	0.648	1.000	
6	HemlockControl vs HardwoodControl	1	0.6019456	6.3330199	0.5135012	0.025	0.150	

```
> |
```



BETADISPER



```
> anova(ants.beta.pa)
Analysis of Variance Table
```

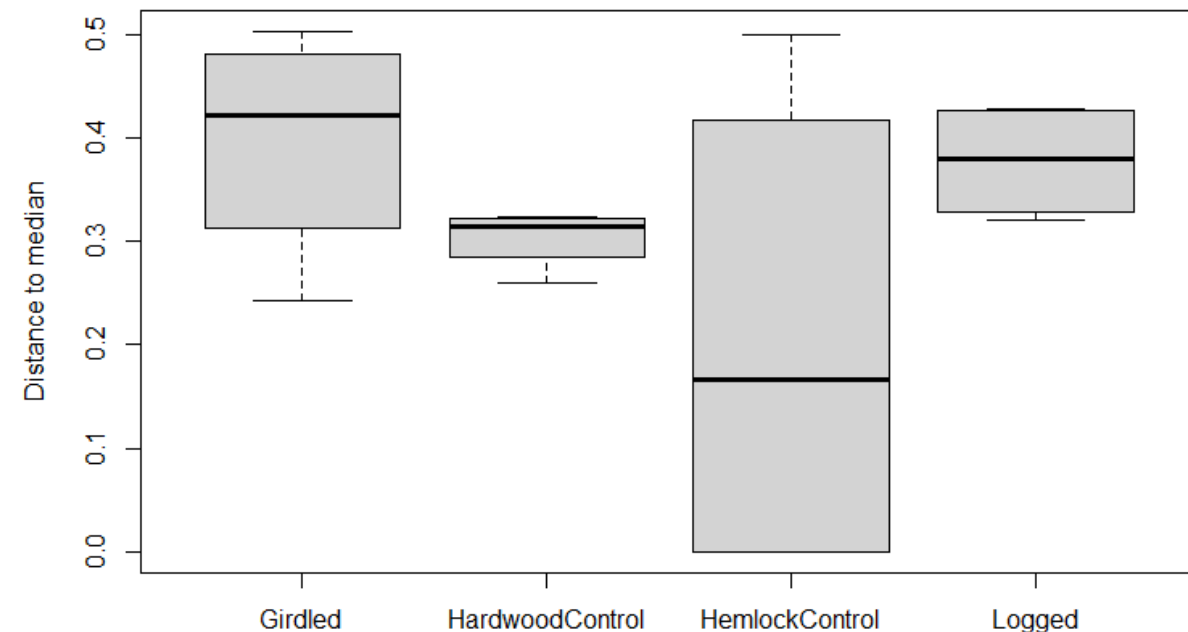
Response: Distances

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Groups	3	0.087628	0.029209	1.4654	0.2733
Residuals	12	0.239198	0.019933		

```
> TukeyHSD(ants.beta.pa, which = "group", conf.level = 0.95)
Tukey multiple comparisons of means
95% family-wise confidence level
```

```
Fit: aov(formula = distances ~ group, data = df)
```

\$group	diff	lwr	upr	p adj
HardwoodControl-Girdled	-0.09392745	-0.3903214	0.2024665	0.7840298
HemlockControl-Girdled	-0.18850468	-0.4848986	0.1078893	0.2831197
Logged-Girdled	-0.01984796	-0.3162419	0.2765460	0.9970601
HemlockControl-HardwoodControl	-0.09457723	-0.3909712	0.2018167	0.7806065
Logged-HardwoodControl	0.07407950	-0.2223145	0.3704734	0.8782710
Logged-HemlockControl	0.16865672	-0.1277372	0.4650507	0.3703064



Principal Component Analysis (PCA)

Used to assess relationships among variables along a reduced number of axes

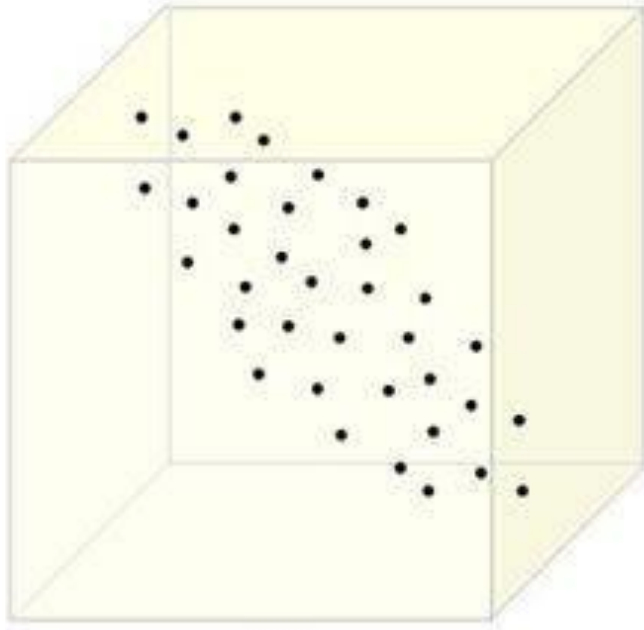
Eigenvector-based method

Eigen decomposition of a dispersion matrix (linear covariances or correlations)

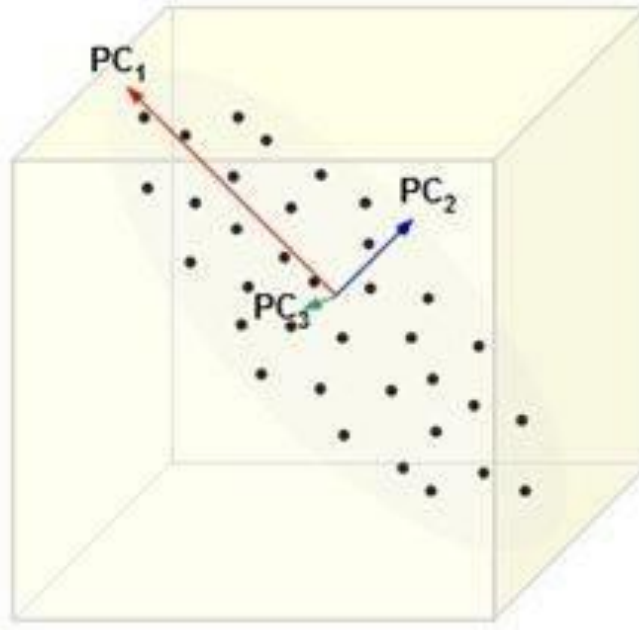
Requires raw, quantitative data

Descriptive / exploratory

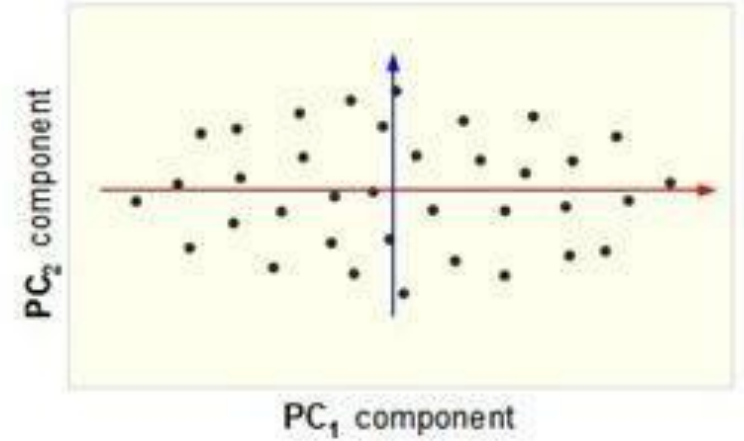
Principal Component Analysis (PCA)



a



b



c

3D



2D

Wide format

2014 & 2015

[illegible]

Principal Component Analysis (PCA)

```
> herb.pca <- rda(herb3[,5:25], scale = FALSE)
> summary(herb.pca)
```

Call:
rda(X = herb3[, 5:25], scale = FALSE)

Partitioning of variance:

	Inertia	Proportion
Total	399.4	1
Unconstrained	399.4	1

Total variance

Eigenvalues, and their contribution to the variance

Importance of components:

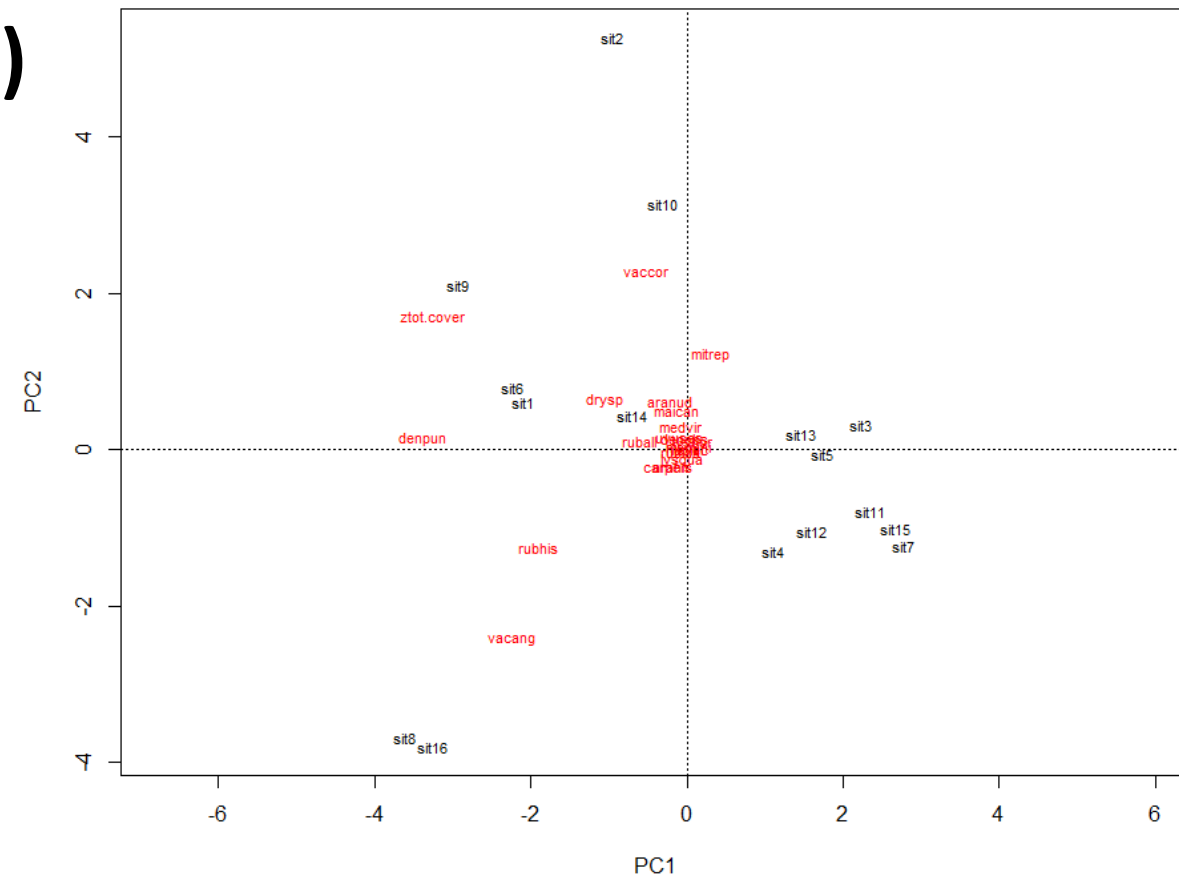
	PC1	PC2	PC3	PC4	PC5	PC6	PC7	PC8	PC9	PC10	PC11	PC12	PC13	PC14	PC15
Eigenvalue	168.4820	95.2045	67.1849	45.0949	11.84576	5.37022	2.68426	1.22225	1.073345	0.510973	0.404821	0.2155915	0.1125490	0.0289611	5.164e-04
Proportion Explained	0.4218	0.2383	0.1682	0.1129	0.02966	0.01344	0.00672	0.00306	0.002687	0.001279	0.001013	0.0005397	0.0002818	0.0000725	1.293e-06
Cumulative Proportion	0.4218	0.6601	0.8283	0.9412	0.97090	0.98434	0.99106	0.99412	0.996812	0.998091	0.999105	0.9996444	0.9999262	0.9999987	1.000e+00

Scaling 2 for species and site scores

* Species are scaled proportional to eigenvalues

* Sites are unscaled: weighted dispersion equal on all dimensions

* General scaling constant of scores: 8.798011

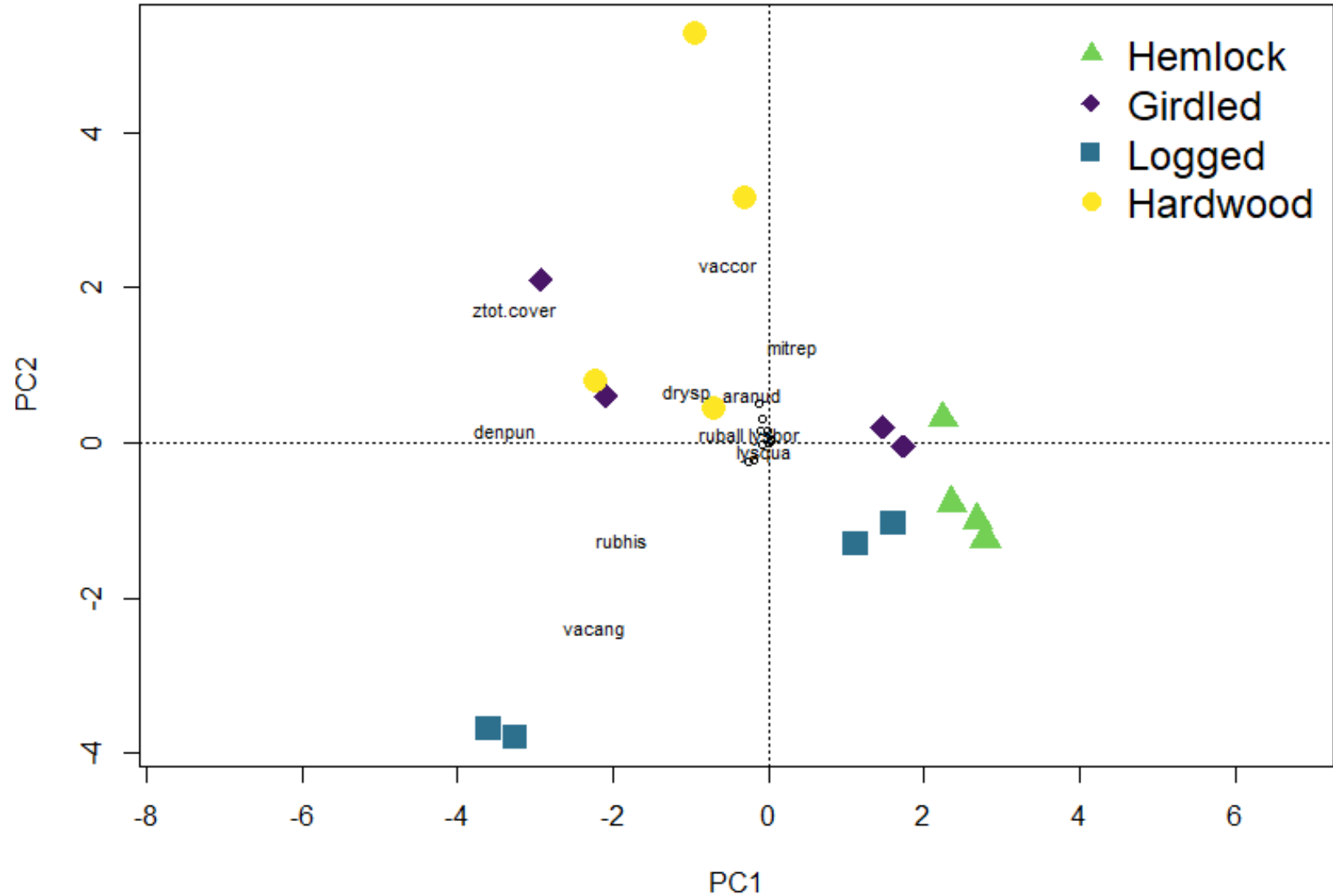


Axes 1 & 2 represent 66% of the variation

Principal Component Analysis (PCA)

Axis 1: 42.1%

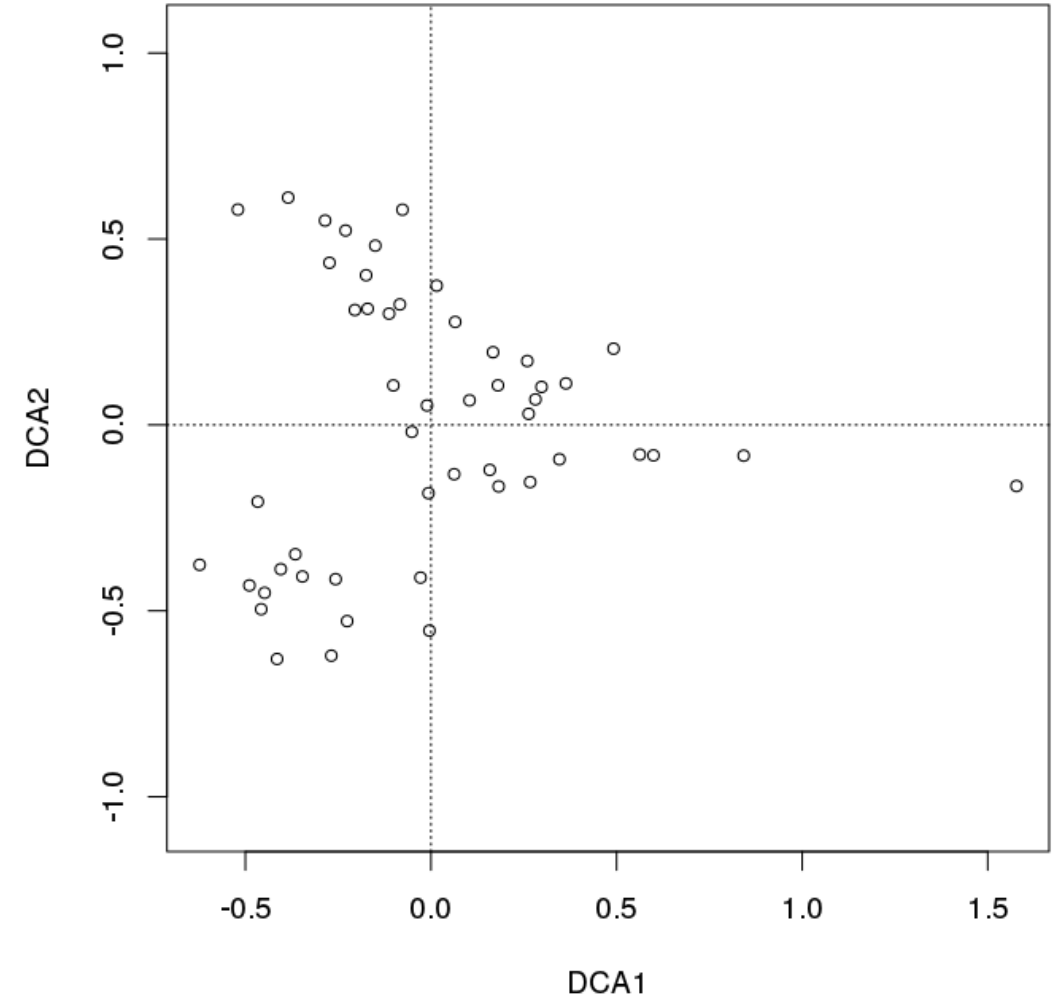
Axis 2: 23.8%



Principal Component Analysis (PCA)

Arch or horseshoe effect – distortion in ordination diagram

- 1) Use sample scores on PCA axes as environmental predictor variables
- 2) Identify environmental variables highly correlated with PCA axes



Types of ordination techniques



R package
vegan

Indirect gradient analysis (aka unconstrained ordination)

- Utilizes only the species x sample matrix
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**Nonmetric Multidimensional Scaling and Principal Component Analysis*

Direct gradient analysis (aka constrained ordination)

- Utilizes environmental data in addition to a species x sample matrix
- Assess whether species composition is related to measured environmental data

**Canonical Correspondence Analysis and Redundancy Analysis*

Canonical Correspondence Analysis and Redundancy Analysis

Used to assess whether species composition is related to measured environmental variables

Explores relationships between two matrices – response and predictor

Combines multiple regression with classical ordination

Can be used to test hypotheses via permutation tests

Linear (RDA) or unimodal (CCA)

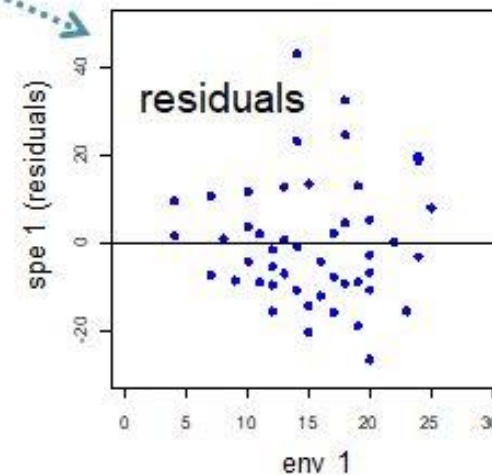
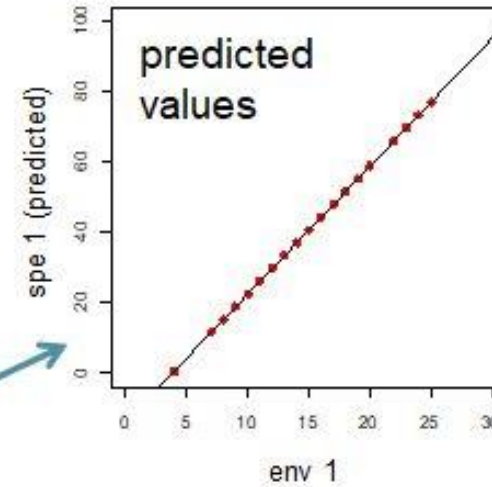
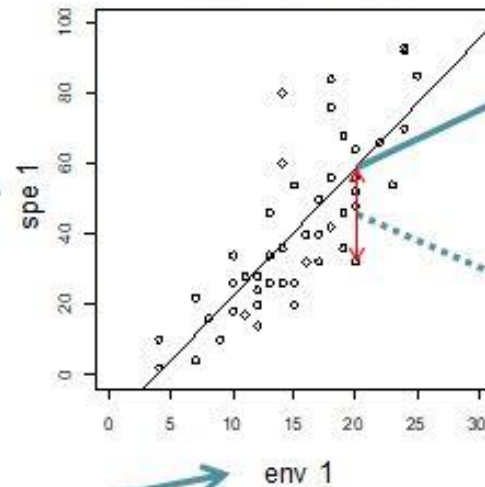
sample × species matrix

	spe 1	spe 2	spe 3
sam 1			
sam 2			
sam 3			
sam 4			
sam 5			
sam 6			
sam 7			

	env 1
sam 1	
sam 2	
sam 3	
sam 4	
sam 5	
sam 6	
sam 7	

matrix of environmental variables
(single variable in this case)

regression of species
abundances on
env. variable



matrix of predicted
values

	spe 1	spe 2	spe 3
sam 1			
sam 2			
sam 3			
sam 4			
sam 5			
sam 6			
sam 7			

	spe 1	spe 2	spe 3
sam 1			
sam 2			
sam 3			
sam 4			
sam 5			
sam 6			
sam 7			

matrix of residuals

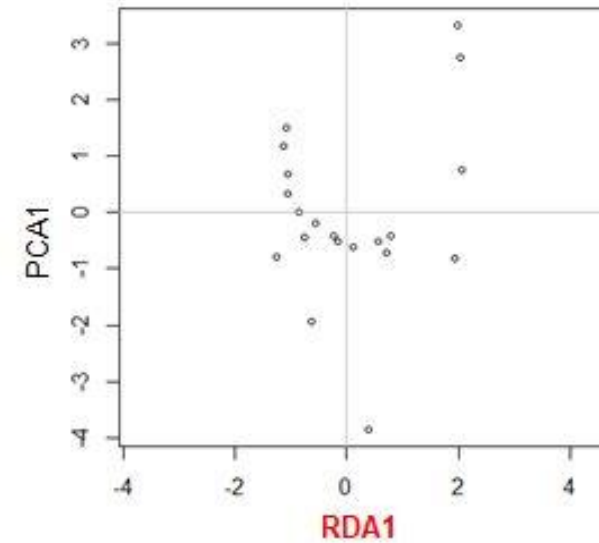
matrix of predicted values

	spe 1	spe 2	spe 3
sam 1			
sam 2			
sam 3			
sam 4			
sam 5			
sam 6			
sam 7			

PCA on
predicted
values

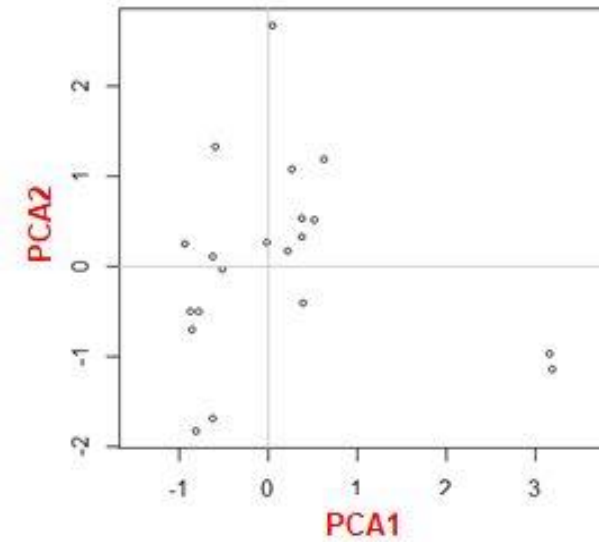


constrained ordination axes



	spe 1	spe 2	spe 3
sam 1			
sam 2			
sam 3			
sam 4			
sam 5			
sam 6			
sam 7			

PCA on
residuals

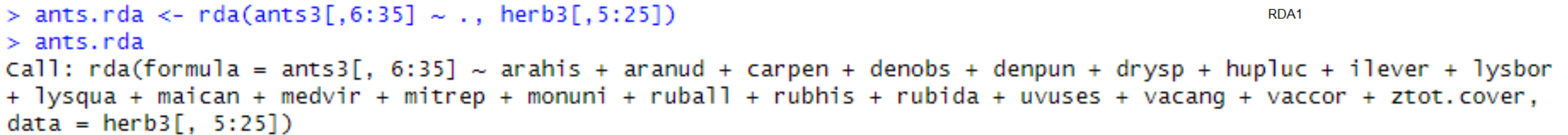


matrix of residuals

unconstrained ordination
axes

Axis 1: 64.9%

Axis 2: 15.5%



Eigenvalues for constrained axes:														
RDA1	RDA2	RDA3	RDA4	RDA5	RDA6	RDA7	RDA8	RDA9	RDA10	RDA11	RDA12	RDA13	RDA14	RDA15
818.0	195.4	83.1	65.5	50.6	18.9	15.9	8.4	2.3	1.2	0.3	0.3	0.1	0.0	0.0

RDA – Hypothesis testing

ordistep()

```
Step: ants3[, 6:35] ~ medvir + ruball
```

	Df	AIC	F	Pr(>F)	
- ruball	1	108.63	4.625	0.025	*
- medvir	1	114.47	12.390	0.010	**

```
---  
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

	Df	AIC	F	Pr(>F)
+ rubhis	1	106.05	1.3543	0.230
+ carpen	1	105.99	1.4067	0.265
+ vaccor	1	106.13	1.2869	0.285
+ monuni	1	106.50	0.9844	0.415
+ arahis	1	106.46	1.0183	0.420
+ denpun	1	106.59	0.9121	0.455
+ ztot.cover	1	106.61	0.8927	0.475
+ drysp	1	106.62	0.8895	0.495
+ lysqua	1	106.69	0.8345	0.495
+ uvuses	1	106.89	0.6741	0.650
+ denobs	1	106.83	0.7232	0.655
+ aranud	1	106.89	0.6721	0.665
+ vacang	1	106.88	0.6800	0.670
+ rubida	1	106.84	0.7152	0.675
+ lysbor	1	106.92	0.6494	0.695
+ ilever	1	107.36	0.3023	0.875
+ mitrep	1	107.25	0.3897	0.885
+ hupluc	1	107.34	0.3213	0.935
+ maican	1	107.34	0.3244	0.950

```
ants.rda <- rda(ants3[,6:35] ~ ., herb3[,5:25])
```

```
mod0.rda <- rda(ants3[,6:35] ~ 1, herb3[,5:25])
```

```
ants.rda.red <- ordistep(mod0.rda, scope = formula(ants.rda),  
direction = "both", permutations = how(nperm = 199))
```

RDA – Hypothesis testing

ordistep()

```
step: ants3[, 6:35] ~ medvir + ruball
```

	Df	AIC	F	Pr(>F)
- ruball	1	108.63	4.625	0.025 *
- medvir	1	114.47	12.390	0.010 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	Df	AIC	F	Pr(>F)
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+ denpun	1	106.59	0.9121	0.455
+ ztot.cover	1	106.61	0.8927	0.475
+ drysp	1	106.62	0.8895	0.495
+ lysqua	1	106.69	0.8345	0.495
+ uvuses	1	106.89	0.6741	0.650
+ denobs	1	106.83	0.7232	0.655
+ aranud	1	106.89	0.6721	0.665
+ vacang	1	106.88	0.6800	0.670
+ rubida	1	106.84	0.7152	0.675
+ lysbor	1	106.92	0.6494	0.695
+ ilever	1	107.36	0.3023	0.875
+ mitrep	1	107.25	0.3897	0.885
+ hupluc	1	107.34	0.3213	0.935
+ maican	1	107.34	0.3244	0.950

```
> summary(ants.rda.red)
```

```
Call:
rda(formula = ants3[, 6:35] ~ medvir + ruball, data = herb3[, 5:25])
```

Partitioning of variance:

	Inertia	Proportion
Total	1259.9	1.0000
Constrained	715.5	0.5679
Unconstrained	544.4	0.4321

```
> anova(ants.rda.red, by = 'axis')
```

Permutation test for rda under reduced model

Forward tests for axes

Permutation: free

Number of permutations: 999

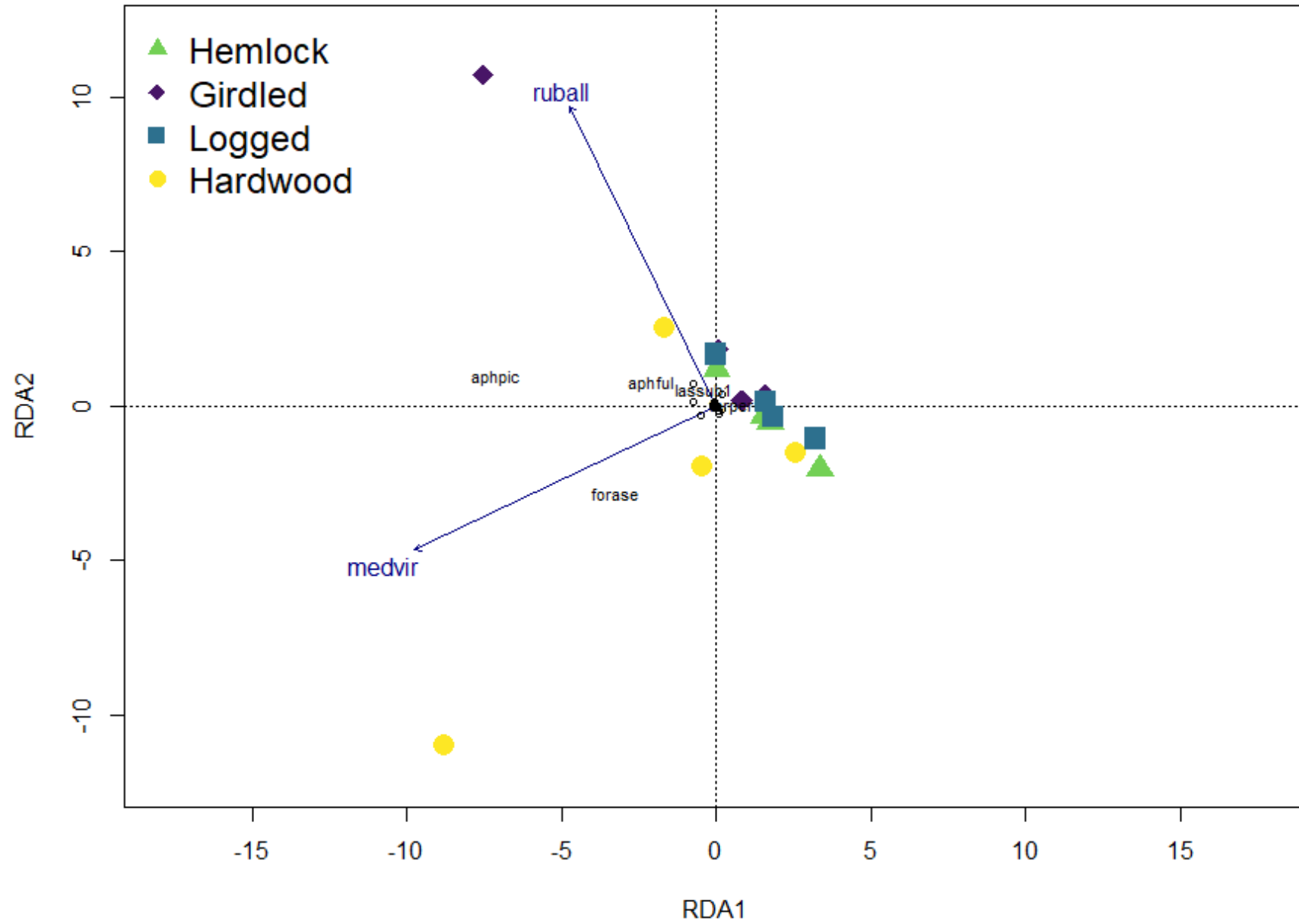
```
Model: rda(formula = ants3[, 6:35] ~ medvir + ruball, data = herb3[, 5:25])
```

	Df	Variance	F	Pr(>F)
RDA1	1	619.03	14.7815	0.004 **
RDA2	1	96.48	2.3039	0.061 .
Residual	13	544.42		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
> |
```

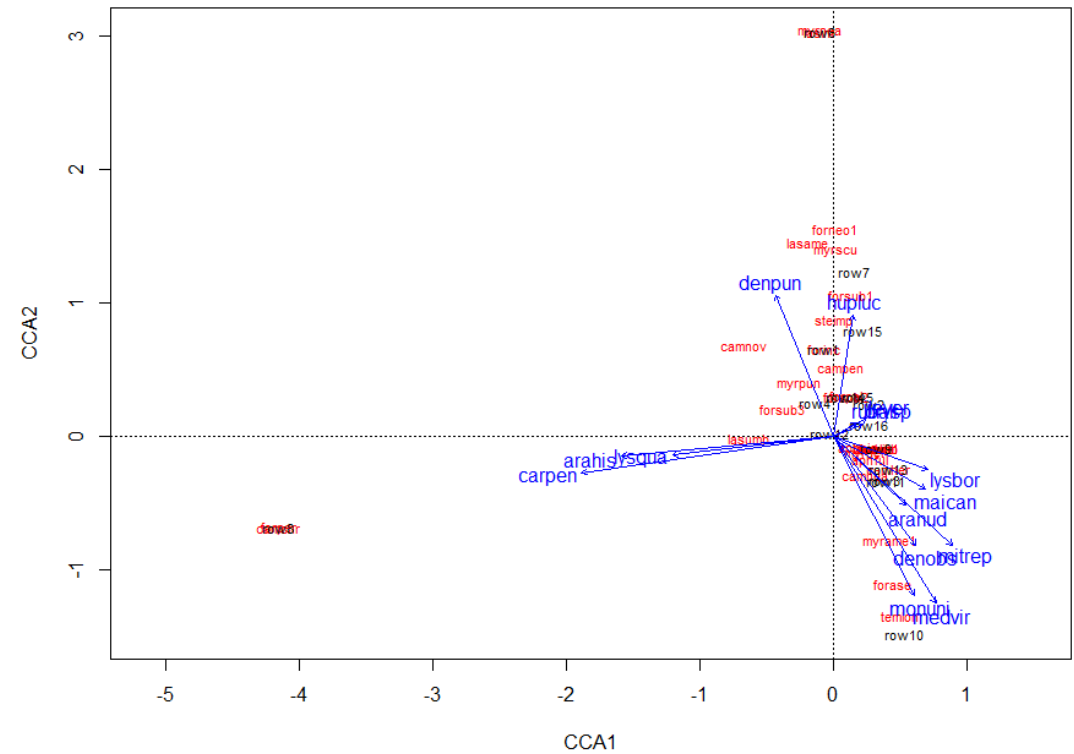
RDA



CCA

Axis 1: 30.3%

Axis 2: 18.3%



```
> ants.cca <- cca(ants3[,6:35] ~ ., herb3[,5:25])
> ants.cca
Call: cca(formula = ants3[, 6:35] ~ arahis + aranud + carpen + denobs + denpun + drysp + hupluc +
ilever + lysbor + lysqua + maican + medvir + mitrep + monuni + ruball + rubhis + rubida + uvuses +
vacang + vaccor + ztot.cover, data = herb3[, 5:25])
```

	Inertia	Proportion	Rank
Total	1.667	1.000	
Constrained	1.667	1.000	15
Unconstrained	0.000	0.000	0

Inertia is scaled chi-square
Some constraints or conditions were aliased because they were redundant

Eigenvalues for constrained axes:

CCA1	CCA2	CCA3	CCA4	CCA5	CCA6	CCA7	CCA8	CCA9	CCA10	CCA11	CCA12	CCA13	CCA14	CCA15
0.5053	0.3058	0.2512	0.1681	0.1206	0.0951	0.0832	0.0618	0.0480	0.0095	0.0086	0.0065	0.0025	0.0003	0.0000

CCA – Hypothesis testing

ordistep()

```
step: ants3[, 6:35] ~ carpen + denobs + denpun + ruball + rubida
```

	Df	AIC	F	Pr(>F)
- rubida	1	72.728	2.2035	0.095 .
- ruball	1	73.374	2.7061	0.040 *
- denobs	1	74.605	3.7220	0.020 *
- denpun	1	75.072	4.1286	0.005 **
- carpen	1	77.799	6.7536	0.005 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*'

	Df	AIC	F	Pr(>F)
+ monuni	1	71.137	1.4601	0.155
+ lysqua	1	71.243	1.3906	0.260
+ aranud	1	71.395	1.2924	0.260
+ medvir	1	71.538	1.2007	0.260
+ vacang	1	71.605	1.1582	0.445
+ uvuses	1	71.866	0.9937	0.470
+ rubhis	1	71.925	0.9574	0.475
+ arahis	1	71.920	0.9602	0.495
+ lysbor	1	72.173	0.8042	0.495
+ vaccor	1	72.226	0.7717	0.550
+ drysp	1	72.126	0.8331	0.615
+ ilever	1	72.476	0.6199	0.660
+ hupluc	1	72.301	0.7260	0.680
+ ztot.cover	1	72.445	0.6386	0.755
+ mitrep	1	72.764	0.4485	0.895
+ maican	1	72.899	0.3693	0.915

```
> summary(ants.cca.red)
```

Call:

```
cca(formula = ants3[, 6:35] ~ carpen + denobs + denpun + ruball + rubida, data = herb3[, 5:25])
```

Partitioning of scaled Chi-square:

	Inertia	Proportion
Total	1.666	1.0000
Constrained	1.098	0.6586
Unconstrained	0.569	0.3414

```
> anova(ants.cca.red, by = 'axis')
```

Permutation test for cca under reduced model

Forward tests for axes

Permutation: free

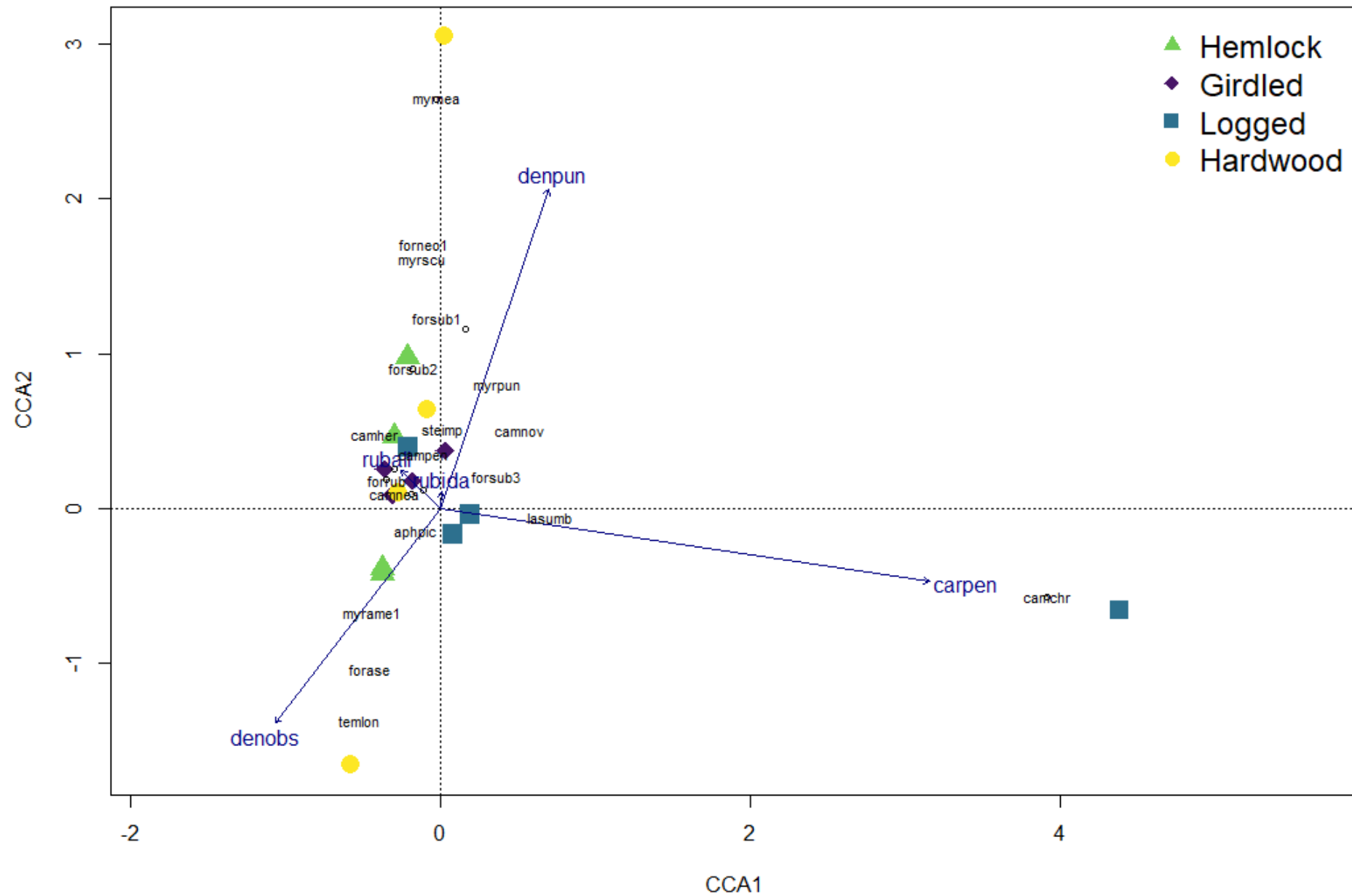
Number of permutations: 999

```
Model: cca(formula = ants3[, 6:35] ~ carpen + denobs + denpun + ruball + rubida, data = herb3[, 5:25])
```

	Df	ChiSquare	F	Pr(>F)
CCA1	1	0.45327	7.9665	0.001 ***
CCA2	1	0.26690	4.6909	0.050 *
CCA3	1	0.21112	3.7106	0.429
CCA4	1	0.10005	1.7585	0.990
CCA5	1	0.06616	1.1629	1.000
Residual	10	0.56897		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

CCA



How do I know which to use: RDA or CCA?

Linear vs unimodal

Detrended correspondence analysis (DCA) to determine the length of the first axis

```
> DCA <- decorana(ants3[,6:35])  
> DCA
```

```
call:  
decorana(veg = ants3[, 6:35])
```

```
Detrended correspondence analysis with 26 segments.  
Rescaling of axes with 4 iterations.  
Total inertia (scaled chi-square): 1.6665
```

	DCA1	DCA2	DCA3	DCA4
Eigenvalues	0.3538	0.1555	0.09320	0.09082
Additive Eigenvalues	0.3538	0.1551	0.08423	0.07622
Decorana values	0.5053	0.1358	0.03636	0.01418
Axis lengths	2.6486	1.5961	0.98833	0.91647

RDA: Axis length < 4

CCA: Axis length > 4